

Software Engineering

Lab 3 – Architecture Selection & Component Modelling

Name: Ramya H R

SRN: PES1UG25CS835

System: Pet Adoption & Shelter Management System

Architecture Selection: We chose Layered Architecture for the Pet Adoption & Shelter Management System.

Reason 1: The system has clear groups of work: applicant and staff screens, application/animal/foster services, and stored shelter data. Separating these parts makes the system easier to understand and maintain.

Reason 2: The adoption process uses several related functions such as searching animals, submitting applications, reviewing applications, updating medical records and managing foster placements. A service layer can handle these functions without putting the database logic directly in the user interface.

Security Advantage: Access can be controlled at the service layer. For example, applicant users can submit and check their applications, while shelter staff can review applications and update animal or foster information.

Performance Benefit: Frequently used application and animal data can be handled by the service layer and database separately. This reduces unnecessary work in the user interface and keeps data access organized.

Components used: Adoption Applicant UI, Shelter Staff UI, Adoption Application Manager, Animal & Medical Records, Foster Placement Manager, and Pet Shelter Database